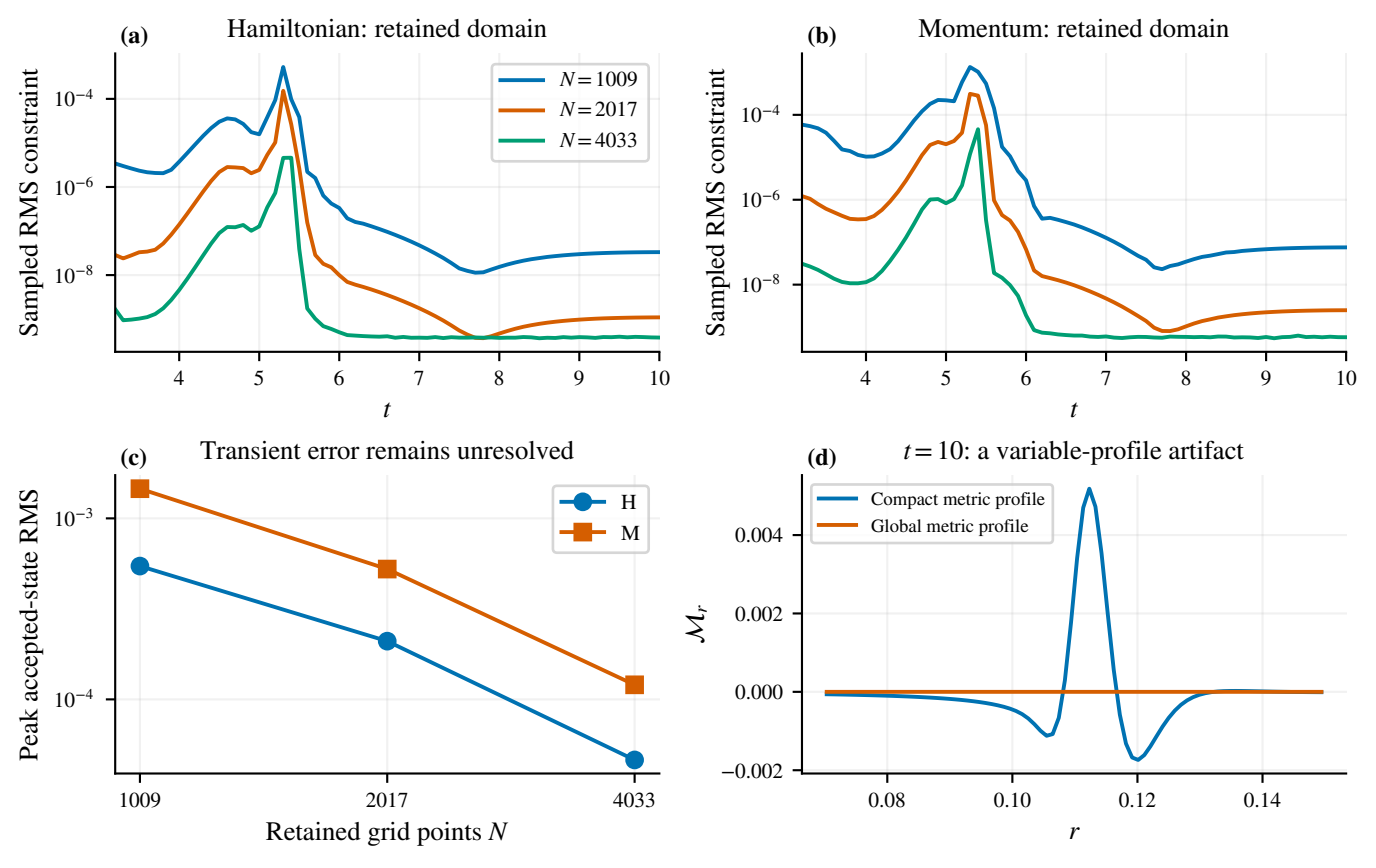
(a)
Hamiltonian: retained domain
Sampled RMS constraint
10^{-4}
10^{-6}
10^{-8}
$N = 1009$
$N = 2017$
$N = 4033$
4
5
6
7
8
9
10
t
(b)
Momentum: retained domain
Sampled RMS constraint
10^{-4}
10^{-6}
10^{-8}
4
5
6
7
8
9
10
t
(c)
Transient error remains unresolved
Peak accepted-state RMS
10^{-3}
10^{-4}
H
M
1009
2017
4033
Retained grid points N
(d)
$t = 10$: a variable-profile artifact
Compact metric profile
Global metric profile
$\mathcal{M}_r$
0.004
0.002
0.000
−0.002
0.08
0.10
0.12
0.14
r